

ActInf GuestStream #009.1 ~ "Collective intelligence and active inference"

GuestStream | 1:49:15

hello everyone welcome to the active inference lab today it is august 20th 2021 and we are here in guest stream number 9.1 with some awesome guests so we'll just start with introductions and then head into a short presentation before a question and answer period so if you're watching live definitely write questions in the chat so i'm daniel i'm a researcher in california and blue i'm blue knight i am an independent research consultant in new mexico and i'll pass it to raphael

hi everyone uh thank you guys for having us over i'm rafael i'm also an independent researcher i have a day job at google that has some but not a lot to do with this topic and i've

been working with pranav and jacob on this topic for about a year and a half now uh it's been a wild ride so right now go ahead

hello everyone uh thank you daniel and pluto for inviting us i'm a phd student uh at carnegie mellon tepper school of business uh

my areas of research are collective intelligence where does it come from building a

causal model social technical architecture looking at collectives as adaptive systems um

you know i was very interested in the uh

active interest framework we met rafael

i and jacob met

you know across different ci conferences

and got this project going so we're very

excited to talk more about and learn

from all the discussions that we have

back over to you guys

sounds cool so let's go to raphael's

presentation and then we'll just have a

bunch of time for questions so go for it

all right before i get started can you

all confirm that you can see me alright

uh yeah we see the slides the slides

perfect all right cool

all right so um

what are we doing here um

so as we mentioned uh

the the interest here is to try to

understand better uh the phenomenon of

collective intelligence and so two

examples here what we're talking about

and they're pretty different examples

one of them is a pretty classic uh

example it's the murmuration of

starlings

top

left i know the bird that's solving a

puzzle on the top right is a crow not a

starling so i'm cheating a little bit

but that's fine uh so

uh the on the bottom the example is the

discovery of the exposure which is an

amazing example of

kind of a large scale long-term

scientific collaboration so those are

both legit uh examples of collective

intelligence and and

yet the way that we approach analyzing

these uh things uh tends to be a little
bit different

so uh if you're doing something around
complex aerodynamical systems or
computational biology uh you tend to
look primarily at the collective
behavior and to be able to do that uh in
a computable way you assume that
individual agents are following simple
rules to react to the environment and to
their peers

and if you're doing something like uh
like anthropology or social science uh
you're probably thinking of collectives
that look more like the bottom and so
you're focused on these really
sophisticated agents that have these um
complex social cognitive capabilities
that are the thing that endows them with
the ability to access collectives and
you try to build a picture of the
collective behavior kind of
bottom up from the some of those
interactions

and both of our valid perspectives of
course but they're speaking pretty
different languages uh and we think that
there must be a missing link here
because if you think about us humans uh
i mean i i definitely feel like i am at
the same time an autonomous an
autonomous individual and a member of
not just one but a bunch of collectives
right so i fluidly go in and out of uh
companies or sports team or uh or uh uh
polity and of course i'm part of my
family

so

all these collectives can demonstrate intelligence behavior that that cannot be explained exclusively by aggregation of of the individual intelligence and yet uh the collective itself is a complex system that is composed of components that are in and of themselves highly autonomous uh complex adaptive systems as well and that's how we frame the the goal of our of our research we're looking at what you call the missing link which is a a plausible theoretical description of the functional relationship between these two scales and uh in particular we think that the the missing link will be found by understanding what kinds of social cognitive features enable the formation of an effective collective within within these kinds of autonomous agents all right to do that uh we've been uh using active inference as a as a framework uh since uh the group here is probably uh way familiar with active inference already i'm not gonna go uh a lot into it uh just pointing out here that we're approaching this uh you might see the language that's a little bit uh different from from some of the canonical stuff that uh

first and team put out we're using some different nomenclature but it's really the same principle we're all looking at an agent as a markov blanket and uh trying to to model exactly what's the the generative distribution that that creates behavior in that in that agent and look at what happens when you put it against some true system dynamics that are that are happening in the environment and of course you you you run the the system as an asian model all right uh accepted in our case it's a multi-scale model right because we're interested both in what happens at the scale of individual agents and the global system for us it's actually quite uh interesting to try to decouple things and look at what's going on in a very specific way modeling many many-to-many relationships uh is really hard so we really focused on the sparsewise subsystems so our our uh model is really composed of a collection of pairs of agents they only have interact with each other within a single pair and we basically compose them the the

what we call the system scale model

bottom up from

copies of these pairs of agents

all right

so

let's look first at the individual scale

where we have um

each fare is operating on the same

physical environment they're sharing uh

you can think of them as sharing one of

these uh circles

um if you're familiar with uh with the

mcgregor model that

has been published in 2015 it's uh it's

a very simple very simple intentionally

uh

stylized uh active inference model that

was originally designed to be

analytically solvable actually so it

allows us to have a really clear uh

view of what's going on

on the system as a dynamical system

and uh we just added some minimal

capabilities in that model so

each of the agents that are marked out

as little can think of them as little

parts within this uh within the circle

environment

they

each of them can only do a couple of

things which is their capabilities they

can only observe their their own

position

uh they can't directly observe the

environment or that that is uh

uh meaning they can't observe their own

position they can only get a clue

uh which is an on-off sensor

generated by a chemical signal on each
cell i'm getting a pin here from a
friendly viewer also known as my wife
saying that my image may be blurred can
no it's good for others
it might look a little different but i
have a high resolution one saved locally
no matter what the final processed
youtube one looks like thanks youtube
all right thanks youtube
uh all right cool so as uh as i was
saying the the main way that agents uh
get information
about their physical environment is by
this uh sensory state which is a binary
uh one on off sensor that's uh you can
think of it as being generated by a
chemical signal or a kind of smell on
each cell and it's emanating from this
from the source position source position
um so that's pretty much it about the
physical environment and agents can go
uh left and right or stay stand so on
that environment
uh now
talking about the relationship between
uh
between agents and their peer on a pair
uh their pure impair is a funny thing to
say but
the
the agent can observe
their uh their peer or their partner's
position relative position meaning i'm
one or two or however many uh
um hops
to the left or to the right they can be
actually in the same position as well

uh and they can also
observe their their motion right there
they're
you can observe whether the the agent
the other agent the partner has gone to
left right or sits on the
last
um last iteration
right um
let's talk a little bit about uh reward
functions uh the agents have independent
reward functions that are two beats uh
with one common peak and one private
peak that the reason why that support is
going to become clear a little bit later
cool
so that's it that's pretty much it about
the individual agents uh as i promised a
really simple uh really simple model
and the global scale as i said uh is a
number of agent pair you can think of
like 80 copies of these two pairs of
agents meaning 160 agents in total or
or whatever and uh
from that perspective of the the higher
scale
the the markov blanket is actually
composed of the
of the collections of all of these pairs
right so they are each interaction
interacting with a
um with a global environment
and uh the cool thing is that we're
we're um
the way that the sensory inputs are
relayed to each agent is as desires so
you can think of it as the top level as
kind of

bringing in some uh some nudges some target states that are that are pretty much the the sort of smells that the agent picks up um all right the other important thing is that the optimal state for the system compo corresponds to the agent's share target only but so from the system's perspective it wants all the agents to be in that shared target but they the agents themselves they don't know it uh why it's important is that it means that there's no exogenous incentive to to achieve the global goal so it's not like just for nudging we're asking all all the agencies the same thing and lo and behold they're going to do it all right and let's talk a little bit about how free energy minimization happens here uh on the on the individual level on the level of the individual agent uh we just assume that there's some process some neurological process within each agent that uh that does gradient descent that's the traditional technique uh but at the global level what we're actually doing is that the behavior that we're simulating at the collective of all these individual agents the the aggregate of that behavior is uh what we want what we would like to see

approximate

gradient descent so that's the thing
that we actually want to
demonstrate here we want to look at the
behavior of the collective
um calculate the the free energy the
collective free energy which is f_{σ}
here in this uh in this situation at the
bottom

and uh see if it's actually minimized if
it's minimized then we uh
then we can say that the system this
this
uh

group of very simple agents as a
collective it's uh performing active
inference at the at the global scale or
not and uh shock review what we find is
that the extent to which the collective
is actually able to do that um the free
energy minimization depends on the
social capabilities that we uh endowed
these simple agents with
um so we focused on two
um cognitive uh capabilities or or
or

features

that are stylized versions of
of things that we
traditionally look at in uh in social
science

uh the

first one is what we call theory of mind
and what we
define that as that is the ability of an
agent to put itself in the shoes of
another and we actually uh try to model
that in a pretty literal way uh so

if you think of shallow versus deep
representations of
of the other you can think of a
shallower presentation as like a
mechanistic oh if they're going this way
it means
it means this rule-based kind of kind of
kind of way which you you could think of
being
an internalization of a rule that i
think somebody else is following
but we actually did hear some something
quite different right so uh it was based
on the this philosophy of active
inference that
uh means that the agent has every agent
has a self-actualization loop and what
we said here is hey
uh the self-actualization loop the
internal part of it needs to be um
implemented by some kind of
quote-unquote neuro neurological
circuits and what if this exactly the
same type of circuit the same structure
were being used to model the partner so
so that you actually also had a partner
actualization loop
so this is quite literally putting
yourself in the shoes of of your partner
right so
uh you have your own beliefs about your
position
and about your desires and you also have
the literally the same uh kind of
belief about
the partner's position and their desires
and you can use that
uh

plus of course whatever sensory or
whatever information you have from the
environment to predict

uh

the the partners actions is exactly the
same way that you're predicting your own
actions the only difference of course is
that uh you don't actually control your
partner so you're predicting that that's
the dotted line from a2

uh out or from a1 out in the in the
right

uh

so you're predicting how you're creating
counter factuals about what your partner
is going to do and then of course you
can observe

uh

observe whether that prediction matches
the

the the outcome in the next round and uh
use that each partner uses that too to
calibrate their model of the
of the of their partner

um and uh

the other thing is you want those two
predictions to be consistent between the
cells right you wanna you want the
beliefs about yourself and the partner
to have some consistency and that
actually helps you uh do what we call a
triangulation

right so you know

uh you know very maybe very little about
your physical environment but you're
seeing your partner do stuff
and uh and moving around and you're
using that information to inform your

your belief about your own position so
that's the the the power really the
power of theory in mind for our model
and we're going to see how useful that
is
now uh the other one is uh go alignment
i now realize that i've been speaking
for a long time so i'm gonna uh
go a little bit faster
and so the other feature that we talked
about is this thing called go alignment
and it's really uh about the ability for
an agent to adapt its goals to to
match another agent's goals and
uh this is a
this is a
bottom-up incentive
incentivization mechanism don't remember
what i mentioned that the system
actually has an optimum but has no
direct way to tell the agents about it
so the assumption here is that to the
extent that you can't that you have
overlaps between the two agents
uh goals and it turns out that those
that the overlapping goals are aligned
with the system optimum that provides a
bottom-up way for
for collectively optimal behavior to
to to happen and we didn't model dynamic
the dynamics of how that go alignment
happens we just assume that it can be
flipped on and off
just again for for simplicity and that's
uh that's the thing that we'd like to
explore further cool
um so we did a virtual experiment we ran
simulations with all four possible

combinations of with or without theory
of mind with or without uh goal
alignment and
and basically what we
and the baseline here which is without
your mind without golem in corresponds
to a scenario where these agents are
pretty much blind to each other they're
not interacting uh with them with each
other at all we can see each other but
they're not using that information
uh and we're gonna see how those
features actually drive the
the behavior of individual agents and
also the system itself
cool so basically what we find is that
uh i don't think i mentioned this before
but we define
in each pair there there's a strong a
perceptually strong and a perceptually
weak agent uh the weak agent is pretty
much following almost uh
random noise that they have very very
poor smell
uh
receptors uh but they still they can
still use the the partner's uh position
motion to triangulate right so
the
the effect here that we see agent-wise
are mostly
what benefit do these uh social
capabilities uh endow the weak agent
with
and
so we see that
the
model with both together actually is the

one where the performance of those the
weak agents consistently improves uh
really well to the point where you know
it gets almost as good at finding an
optimum as the
as the
strong agent and
uh why is that so the the basic
intuition is that theory mind by itself
uh in this model is not able to improve
outcomes that that much because
um remember each agent has two uh two
goals which correspond to the
the two peaks here in the in the
middle drawing on the left side and uh
because there's this this great
ambiguity right if i'm going to
to assert in a certain direction uh what
goal the
it's hard to interpret that in terms of
going to a specific goal if you don't if
there's two possible goals where you're
going so
that limits how much useful information
the agent can get from the
from the
strong
and uh goal alignment itself uh doesn't
help by itself
because now you have a single reward
target target but you you're still not
uh super well defining where you're
going but putting the tape up the two
capabilities together actually does
result in a grass skin improvement so oh
i'll let it let that sink in for a
second uh theory of mine knowing put
putting yourself in the shoes of another

plus uh having a goal alignment the
capacity to to
to
share a goal uh it actually creates
benefits for the the agents that are
compensates helps a lot to compensate
for deficiencies in the in the
in the agent's ability to navigate its
physical environment
and uh
that also is true uh looking at the
collective so remember we were talking
about whether these uh models can
minimize the the collective free energy
f sigma so
we're interpreting this uh these runs of
uh
all these runs of
optimization for the
individual agents as part of one single
uh bayesian inference step for the
collective right and what we see here is
uh the
if we're doing this right the graph
should look like uh
like a gradient descent down the the
free energy gradient and
this only really happens consistently
towards an asymptote of zero in the
fourth model where you have both of
these social capabilities as we
mentioned you know get some improvement
uh you you have some abilities from
these agents that are coming from their
individual capabilities but when you
when you put uh the social cognitive
together with the with the
uh capabilities about the environment

that's when you get the ability to
really minimize the the system free
energy

all right so that's what uh what we uh
found in our paper and we think it's a
pretty interesting and compelling uh
example of being able to link uh still
in a very stylized way but
uh but very clearly link uh social
cognitive capabilities at the individual
level that are just that are not just
simple rules right it's not about simple
rule following but it actually has to do
with interpreting the other and using
that information

and linking that directly to um
to active inference at the collective
level

uh and where we'd like to take that i
think the the four most exciting uh
things that we've been discussing the
last couple of months are
first of all

uh can you actually
look at experiments with with humans and
uh

because you know these capabilities
these software

these social capabilities are things
that you can measure uh through your
mind and the ability to align goals for
things that you can test out with human
subjects and uh can you

use our model to to

to

represent and simulate and contrast with
what happens in groups of of humans and
see if

the
differences in their abilities to
to
to have theory of mind and to align on
goals
matches up with with what we're defining
here as free energy minimization that's
one one idea
another one is you know collectives one
of the things that collectives can can
have is behaviors around shared
resources and these resources endowed
them with capabilities in different ways
so that creates we think that creates
sort of an on-ramp to understanding the
the idea of economic value
um
that's we've been joking and jokingly
calling it for tsunamics
this is uh
the further down you go the more
speculative we get but we think that
uh
active inference can actually help us
understand this notion of
of where does value come from is it
really just uh
related to scarcity or is actually also
related to to the capabilities that it's
endowing with different agents how these
two things relate to each other so i
think that's a there's a there's a
powerful idea there
um this notion of uh gynomics uh i wrote
a little bit about that i think it's uh
it's been
starting to pick up scene in some areas
and uh you know ultimately the

uh all of us are embedded in one big collective which is gaia from gaia theory uh it's the planetary scale uh collective intelligence that some sometimes works really well sometimes it doesn't work really well can we use these sorts of ideas of multi-skill um active inference modeling to to improve how we define and model uh the the behavior of that guy assistant finally going from the pretty big to the pretty small individuals and individual humans brain uh we've been looking at jeff hawkins's uh a thousand brain theory from his recent book and there's a lot of interesting ideas that look a lot like active inference but uh as far as you know uh nobody said try to model what what he says is the fundamental unit of intelligence which are these cortical columns in the new neural cortex nobody tried to model those using active inference and as the name implies the the the main thesis of this theory is that the intelligence in our brains comes from the collective behavior of all these all these cortical columns interacting by a certain algorithm and so what it would look what would it look like to try to implement this uh this year in the context of of the brain and what inter-protocol agent capabilities are required

so

i'm going to pause there i've i've

overstayed my welcome

i'm really curious to hear what kind of

questions or or thoughts

we have

in our homes yeah

thanks

awesome thanks for the fascinating

present you're muted then

thanks sorry um

great presentation so maybe we could

start with just uh pravnav and then blue

just

what is bringing you to being right here

and what are your first thoughts either

being involved on the work or on the

collective behavior side more generally

blue are you going start like

author's profession

thank you thank you uh

no i think uh

so this work in particular

uh like i'll start with this work as a

context to set up the

the bigger idea is

a part of the thing that we

i'm interested in we're interested

together

is figuring out

one

understanding causality of where does uh

you know collective intelligence emerge

from like what are the interactions

between humans or humans and machines

those things uh while you can have

causal theory from a management social

science perspective

it's very hard to know that it is optimal like is that the outcome uh like a statistical you know mechanics approach to looking at ideal gas behavior and saying that oh the at the aggregate level this is how the gas should behave but you can look at the micro interactions uh underneath so i think for from my perspective active inference is like you know this may not count correct but from a social scientist perspective it is an a theoretical way or physics way of looking at uh the markov blanket like looking at the statistical mechanics without worrying about how exactly does theory of mind come about so as raphael explained we had a partner actualization loop we think the analog in social science is theory of mind uh and and what we start to demonstrate is that even if there are individual uh active inference agents they will need some form of partner actualization and in humans you can observe that as the capability of theory of mind to be able to you know use each other as sources of information so if i cannot uh work in my environment because my physical skills like looking at you know smelling the uh different chemicals at different levels is low but my social perceptiveness is high so i can look at

my partner and infer something about my environment then i can still work in the environment right but that alone is not enough for us to work as a collective so the free energy of the collective will not go down just if everybody can understand each other's perspective or step into each other's shoes so then we need something more so the second thing that we introduce and you know test out is some form of aligning goals like there has to be a reason to find you know common peaks uh again the current model does not say how it comes about that is a social science type of question because the causality like do we trust each other do we have a good communication processes or not like there will be accuracies and efficiency issues in communication themselves all of this is not you know aif doesn't worry about that but it tells us that these two capabilities at the individual agent level are important to get a collective outcome like i think from from our perspective that is really the uh core contribution that we are saying that there are some cognitive capabilities that are beyond an individual aif model the way mcgregor had set it up that are needed for it to show up as a collective all right there's much more to be done here but we are starting to draw these patterns so the exciting thing that uh raphael started uh you know talked more about was can we map those which is uh

experimentally if we run
uh human agents in a similar environment
and you know and ask them to do a
similar task and we have aif agents
working together
can we create a mapping of what is the
you know theory of mind index like the
score for this person for this person on
this scale because that would help us
preemptively say that all right if you
have a collection of people with this
distribution of theory of mind like
their ability to uh understand each
other's like social perceptiveness which
is a standard scale uh in social
sciences
i think it's called rme reading the mind
and the eye that's a standard scale so
like if we can create a mapping from
a real uh indicator of theory of mind
onto these things we can start having
some predictions that this team is
likely to succeed more than this
combination of people right so that
starts to open up the area all right now
we have some interesting uh
uh understanding of how the whole system
works and can we draw it to real human
so like this is very much the beginnings
of a
grander uh
research statement and there is like
research direction that we are trying to
pursue and using aif
in uh tandem with
you know cause and theories or uh
management or social science theories to
figure out more that how can we you know

help teams better or design teachers
does that like
is that enough like i didn't go much
beyond that but like very happy to talk
more on other kinds of words
over to you
thanks that was great um yeah i really
enjoyed this paper from um my background
is like biological sciences and so like
i've done cellular and molecular
neuroscience like all the way through my
phd until i kind of took the jump off
the diagonal um so for me it's i'm not
as well versed in the social science
aspect but i think
a lot about collectives and i'm super
drawn to collective intelligence and
even um hierarchical systemic
organization right like so how you know
bodies are built right from like
molecules to cells to tissues to organs
etcetera um and even like to societies
and ecosystems um like the gaia theory
that you were talking about earlier so
so this paper was awesome for me i do
have like quite a bit of questions um
especially at like the social science
front like in the beginning of the paper
you guys outlined like a whole lot of
different um
you know potential ways that we operate
like as
multi-system um
multi-agent systems and so like a lot of
them you have theory of mind in there um
and also the goal alignment but you also
had like shared norms and folk
psychology and so what what made you

choose just those two as opposed to like
testing all the other ones out or did
you try others or did you not figure out
a way to like mathematically get it
together or what was your logic there
expedience

right now do you want to take that
uh

yes i think so we had
about six months of discussion of what
would be the scope of the paper and
that's where uh because it was like so
many things like as you can see here i
found this and we are all very excited
about looking at it at the level of
christian

but before we could jump on to any of
those things i think it was important to
boil down as to what is the key
you know pieces that are needed before
it can go to specifics so the key piece
was

at this point aif

does a very good job of explaining like
an individual interacting with an
environment and how it operates
so

yes we can go to norms yes we can go to
these things but these are details about
how does you know norms emerge and at
the end of the day again i could be
misunderstanding some pieces of it
because i i'm coming from social
sciences to active influence so pardon
my uh pardon me errors there and please
correct me my understanding is that aif
does not give you a sort of a human
level explanation it is the same kind of

collective outcome that we would see if
of cells clumping together we would not
call it theory of mind in cells because
i don't think we have a
feature that says oh cells have theory
of mind of other cells like it could be
like it is still an analogy so aif is
not telling you that this is the
mechanism it's saying that you need this
as a functional piece to it so that the
energy minimization happens and that is
demonstrated to the map so the reason we
did not go into a specific
implementation was because i think that
would be the next you know set next set
of steps we first need to say that
at the abstract level at the you know
baseline level concept level we need to
have certain additions to an aif
framework when looking at collectives
as against uh so one way of looking at
collective that we you know
do in the systems free energy system
level free energy is that we look at the
entire system
as a
you know aif
thing and plots free energy but that is
very different from being able to say
that oh how are these two members or
these 15 members within this system
interacting with each other what are the
capabilities that these people have that
is different from this whole system as
an you know active inference system
and that that is where i think if i am
not
like i don't mean to complicate this uh

or sound vague i think we first wanted
to establish all right what is the
minimum mechanism that we need if we do
that in one paper at the simplest level
then let's start digging into how does
the goal alignment or norm uh formation
come about so that can then be a
parameter that certain
teams or certain set of individuals or
subgroups of individuals are better at
forming norms while others are not you
would expect these people to minimize
their free energy as a subgroup better
than this other subgroup so that would
be an exciting extension in which you
can go
but we didn't want to jump there because
it would be doing too much
at least that's where we were at raphael
is that

accurate or did i completely

[Music]

from my side where where i
where i got to to the beginnings of
what we were discussing is i had this
question in my mind of
you know it takes a ton of cells to
create a
great collective behavior
it's usually
it's usually a pretty big discrepancy
between the the different scales at
which you know different levels of
hierarchy are are
are
are set up if you think about a
cell and then uh
an organelle sorry an organ or

clearly not a biologist but
uh even in um
in these examples that we gave with with
starlings or other
examples with
other animals other than humans it looks
like it like the
the the behaviors of ant colonies and so
on it's a it's a collective behavior but
it's really driven by the the
the statistical mechanical uh
implications of of these relatively
simple
uh rule follow group following behaviors
whereas
the
sword i don't claim to have demonstrated
it so i'll continue calling it a hunch
the hunch that that we had is
because humans have this more
sophisticated
uh
set of abilities
that
you know
implicat implies some uh some
some ability to
to
do things in a non-rule following way
meaning that they're not uh they're not
cause effect relationships at the at the
uh individual level uh that
allows collectives of humans to be
formed in a more uh
in uh with fewer humans and with more of
more of a fluid
aspect
and meaning they can the collective can

also have more flexible behaviors
again don't claim to have proven it but
that's where
i i i feel like the
um the two the two capabilities that we
that we picked came from because theory
of mine
true what distinguishes true theory of
mind from just having a heuristic about
how your partner is going to behave is
really this ability to to
to
put
put what you're seeing uh in the world
through the same
the same active inference loop as you're
used for
for yourself which will call putting
yourself in the shoes of another and
that's like inherently not uh
not a simple cause effect thing
the
uh the other one with goal alignment
even though we didn't demonstrate it
quite quite as as deeply i think it
follows the same principle of you know
it being this kind of like mutual
adjustment of expectations that
eventually gets you to uh
to uh to a shared goal
and so you feel like that
that has the kernel of an idea about
what makes human connective special
awesome
one comment and then i'll ask a question
from the chat you kind of pointed out
that there's a lot of modeling at the
single agent level and then sometimes

when you have a massive number of agents
interacting you can almost take these
mean field approaches or
statistical averaging and then the kind
of complexity is in the middle where
there's small groups and reconfiguring
so that was really interesting
and also just the challenge of different
words used for the same phenomena across
scales or across systems
i understood it as sort of like you were
saying this is the continent of theory
of mind these are some of the key words
or the questions and then that's just a
first connection and now you could
reconfigure what the collective could do
or do other experiments
so here's a question from the chat
they wrote a feature of the model is
that order emerges through the bottom-up
self-organization rather than top-down
priors like most active inference models
on morphogenesis
however the goals were pretty much
hard-coded into the model what
self-organization did is only to select
the collectively optimal decision among
a set of possible decisions
so i'd like to know more how the model
improves on existing literature and
importantly what it says about
intelligence which is understood as the
capability to understand the world
rather than just optimize outcome given
a set of possible decisions
all right again
yeah talk a little bit to that and then
if you have anything to add

um i think this uh
this really points to to
another limitation
presented by time when we decided to to
um
to publish this for for entropy we had
this
this notion of being able to investigate
how goals emerged at the collective
level as well and
uh that turned out to just not be
feasible
in our time frame but i totally agree
with the critique that
ultimately what you do want to
understand is is also how goals emerge
we talk about that a little bit in the
paper i think the the one one possible
approach is this notion of
of
um
exploring states and uh as a collective
and uh
and seeing what happens to individual
uh outcomes and if if there is an
evolutionary nudge that would that would
that would align incentives naturally
you know from the from the perspective
of the
um
of the various combinations of all
of
individual incentives as well as if you
think about social incentives they have
these ideas of keeping up with the jonas
jones's or or however there's a bunch of
ways to
to

model that
and we think that given uh
given that kind of
evolutionary nudge towards uh go
alignment we could see that emerge as a
as a
totally bottom-up uh model as well even
having said that i don't think that the
model is uh completely top-down uh given
you know
it's not giving like
so such strong constraints uh and uh
the
the way we try to to
to
tease apart that that thing is that this
notion of everybody has two different
goals and uh but the system has a has a
single
single optimum state
and but it has it's not like directly
telling it's not directly giving
incentives to the to the uh individuals
to to do it right it's just like quote
unquote by coincidence the the
the
shared goal asc
is the
is also
the
the optimum for the system
of course you know as we then as we
showed if without a way to align on that
shared goal it's just as likely that
each
each uh individual will just uh pursue
their own uh private goal we did play
around with some models where the the

shared goal was actually a lower peak
entering the reward function than the
private goals and uh of course that
meant that that
absent goal alignment
the agents were more likely to pursue
the the
the private goals and the shared goal
and so you have a little bit of that
flavor of okay i think we can
uh we can at least say something about
that uh
about that relationship that is not just
the traditional bottom-up composing
goals and of course the units are gonna
are gonna
strive for us as well
you have i think i'll jump in and add a
couple of oh mostly
how we arrived at this visions i think
so several uh uh like i think great
question this is the kind of we started
with a complex model and
you know evolved it into what's the
simplest instantiation for the goal of
this paper so let me walk you through a
couple of different things
which
we started out with but we're now
planning ahead on so we first started
with the idea that
okay if you're looking for a minimal
mechanism we need something through
which and i'm talking in analogy in like
you know two agents teams terms not in
uh aif terms uh we need the two agents
to not only understand each other's
goals or distribution of desires

but also compile them
right so the the simplest way we
initially started with that we gave each
individual their own twin beat or three
beat or you know how it however peaked a
distribution of goals in their
environment and said okay you can just
take them and combine them
but what we started
encountering in that piece was that
should we be thinking about
having an additional value
for approaching shared goals that is
when the peaks are combined if my
partner also has this similar peak it
should be given higher value so there is
like there is a synergy bonus or there
is a collective goal bonus that should
be given that minimizes my uh you know
selfish desire to go after goals that is
only mine like my private goals so like
we've played around this
sort of decision space but we found out
that i think we'll have to create
another parameter on giving them how
much is their preference for personal
goals versus preference for collective
codes so that's an additional parameter
that can be added to the model to each
individual that they have
uh i i forget the name that you're like
selfish buyers or like team buyers
something like we created something of
that sort uh and that would determine so
you could also end up in a place where
me as an agent is always going after my
personal goals but rafael as a team
member is always going after team goals

or shared goals like that would be the distinction that we'll be able to draw out

that was not part of the scope of this paper that's why we you know removed away that's a thing and simplified it as everybody has only two peaks and it is very clearly known that this is the shared peak and this is the unshared to simplify the model and the explanation uh what you can do is make a general version of it that you can have a way of compiling them you can add on a parameter which can say that i have a dream team preference like i have the social science equivalent would be my collectivism orientation uh

individualistic versus collectivism

collectivistic orientations right like

if i am very collectivistic i am very likely to choose a goal that is good for the team

if i'm individualistic i am likely to choose a goal that is good for me right i'm not optimizing for the team so like those kinds of things can be built in and

hopefully we'll be able to explore that further but we realize that that orientation by itself is not

required to minimally explain the emergence of characters right so one interesting piece again not explored in the paper was

over here

the idea was that both

agents should end up at the shared peak

so they are trying to seek the common
peak

what if

uh

so that is how systems high system
performance was uh defined but what if
it is a

what you would call a disjunctive class
that is there are multiple goals in the
environment the team as a whole should
hit all the goals as against compile on
a single goal what does that look like
that is cooperation to achieve all the
goals

our model currently cannot handle that
it does not know how to coordinate
separate goals that are contributing to
a team over here the assumption there is
that

synchronizing or converging onto a
position is what is good collective
behavior that is not true like as you
try to

you know do more analogies on which
direction you want to grow the model
that would be an extension that one has
to think about that

how are we defining what you know better
collective outcome looks like like that
depends on the environment you're
working in depends on how the goals are
broken down so those are the places that
we would like to go so you're absolutely
right that it

it is essentially in right now an
optimal decision-making
situation because the simplest feature
the goal was not to make the

optimization the complexity of the paper

the goal was to say that you need a

heuristic

to find the common ground that's it and

here is the simplest one that we

presented

does that make sense

i think it brings us to this question

about what does make the higher order

group

an active inference agent versus really

what does make something specifically

an active inference agent at any scale

is it just enough to have input of

information and policy selection isn't

that just control theory more broadly so

how do we um

figure out who is and isn't the active

inference agent at the level we designed

it at explicitly which was the lower

level formally defined and then

where really your claim is kind of

building the bridge to which is

statements about collective phenomena

and collective behavior so maybe either

the authors are then blue

yeah and uh i'll i don't have a ton to

add to that i just wanted to say that

you know but i'm not i don't have a

strong background in complexity to

theory or or even in dynamical systems

but

i was

i was reflecting on on those sorts of

things and it struck me that

this assumption of gradient descent

uh you know the assumption that the

optimization is is even possible

is
is such a kind of an ingrained thing
when you think about
this kind of modeling that
exploding that assumption a little bit
and just
and focusing on on exactly this question
of is it even possible for a for uh
for a system that looks like this to
actually like in in practice uh
minimize this free energy functional
right is it
and uh i think it it
again my day job has to do with some
some issues of organizational dynamics
and so on and i think that has a lot
actually to do with with
uh system dysfunctions because
the
the
basic assumptions of
of
of
what needs to happen in order for things
to fall into place is
or sometimes you know not trivial at all
and uh can you know the efficacy of a
team of the of the collective at
uh
doing that kind of of minimization which
we really understand as achieving some
kind of shared interpretation of the
of the world is
is really what's at stake here i also
want to add really quickly that uh we
cheated a little bit our system is not
an active the the global system is not
an active interest model it's just an

interest

uh model uh it

again came down to expedience

so it's the the collective as a whole is

not acting back into the

into the

uh environment so i think

there are bigger questions there around

how does that

bigger loop evolve and how does that

influence the capabilities that that

agents have right if you're like really

trying to just to get your system to to

act optimally on the world and the

levers you have are the kinds of

of social cognitive capabilities that

individual agents have would that select

for more

for

more agents with more theory of mind

with more go alignment with the mix

so that's the kind of things that we

would expect to see with the broader uh

dynamical lens at the

system scale

thanks blue

yeah so um

from the like complexity

uh perspective and kind of piggybacking

off of like the bigger system and what

is the bigger system so

i don't know if you're familiar with uh

maybe integrated information theory of

consciousness but it really looks at

like to what degree the whole is more

than the sum of just its interconnected

parts um and then um there's there's

that like angle but then there's also a

recent paper by david krakauer that
talks about the information theory of
individuality
and they really talk in this paper about
how in order for a unit to be a new
individual um there has to be like so
and this goes even back to like the
origins of life theories right like so
you start off as like a collective of
whatever molecules and then what makes
you form a cell right and so what has to
happen in the integrated information
theory uh paper um or sorry the
information theory of individuality
paper is that there's this
bi-directional information flow from
both the bottom to the top and then from
the top to the bottom which you also see
like in human organizations when like
you get a group of individuals that
align on some kind of collective goal
like so somehow
a leader emerges right like so so if
everybody's kind of like well
i don't know what to do and like you
know how people are they act like that
so then somebody will take charge and
direct the group even in an emergent
unstructured way there becomes this
bi-directional information flow that
makes something a collective unit and
and i just wonder um
like i i just find myself questioning
whether you have that if you've built a
new collective unit or not so and do you
think you could and
can i jump in
quickly so

you bring up fantastic points i think
the way
i have simplified and understood active
inferences that you have an agent it is
considered to have an active uh
working in an active infrastructure
paradigm if it is interacting with its
physical environment to reduce its
surprise right like it is trying to
act on it build a hypothesis and then
act on it as simple as that
but when we try to move to collectives
what we are changing here so like when
uh so for example if you have to say
is this single human an active inference
agent we ask does it interact with a
physical environment you know to
optimize an outcome then we ask is this
organization or a team
acting as an active inference as a
system you can say that oh this
organization is interacting with its
physical environment with the resource
environment and doing things
the distinction that we're trying to
build is how do you go from here to here
which means when there are multiple
individual aif agents in addition to
having a physical environment you also
have a social environment
and
that is where
you you have to be interactive right so
like now you are so the idea of theory
of mind or goal alignment all of this
results in the like from my perspective
in the social environment place which
you know leads to a system as a whole

behaving such that it looks like it is
uh you know an active inference agent it
appears to be an active internet agent
because it is able to
very dynamically work on its physical
environment but coordinate or combine
its information that is coming from
decentralized sources internally to get
you know the you know minimum free
energy
type of action so now building from like
looking back from there the idea of so
we don't do in our current paper i think
as rafael said we are not modeling the
entire system as an af agent what we are
saying is this is simply a statistical
look so if you look at all the different
runs and compile them together it would
seem as if
in most cases the behavior of the system
is such that it aligns with reducing
free energy it is not actually actively
doing things this is just a compilation
or aggregation of the different you know
simulation runs that we've done
so
because you're not trying to build an ai
model at home because at the whole level
if you're trying to build an af model
the model would look different you would
not have separate individuals in the
world uh they will have to have you know
act as uh some somewhere you know joint
at the hip in which there is perfect uh
or there's some form of information
integration going on so the the idea
that i think you brought up
was

that in an unstructured environment
there's a leader developing and they you
know bring it down so this
fascinating idea and i think the way we
were thinking about this was
if we are going down the direction of
when goals are merged do we have a
bonus for
convergence right where does that bonus
come from so i think one way of thinking
about this would be that that is the
leader's job the leader's job is to
change the weightage that each member
gives
to you know pursuing a common goal so
if left to my own devices just you know
carrying the previous example i might be
very strongly focused on my individual
goal that is not shared with it but if
we have a leader amongst ourselves the
leader is changing my the way i weight
these different uh goals in my
perspective and if they're able to
convince me that the common goal would
be a better outcome for you so like
in in my type of theorization i would
say oh negotiation and goal alignment so
that would be that we are talking and we
are saying that all right let me
understand what are you after are you
looking for status are you looking for
exciting job opportunities are you
looking for uh you know more higher pay
let me understand what your underlying
desires and motivations are and let me
try to get them
closer to what the common goal is like
by your pursuing the common goal you

will get x and that motivates you and you know from from like a very modelling perspective it changes the weightage that you assign to common goals and that kind of individual interactions

i would not say that it doesn't have to be a single leader it could all be decentralized that people are helping each other understand why you know following the common goal is better for each one of them that is a discovery process in that sense it is you know very directly looped that once a collective starts forming or once consensus start forming it becomes easier and easier or more more profitable for other individuals to join into you know the group goal because there is very highly high chance of the success or some other resources that you know you get as a payout like those kinds of directions we can think about doing again none of this is done in this model it's a fairly simplistic model to talk about

let's think about social environment how do you go how do you bootstrap from aif in a physical environment to what do the agents do such that it looks like at the high level it is aif it is not at this point but minimum cognitive architectures

can i ask you for power yeah yeah um so

right like in the global system that's not an active inference agent but like each individual agent is an active

inference agent and then they form this
like dyad so is the dyad like this meso
at this meso level is the dyad an active
inference agent also or no

[Music]

i think uh raphael feel free to
you know

disagree with me i don't think it is an
active inference agent at the diet level
because there is no

uh

there is no changes that are happening
in how i infer about my goal states
or how i like the only thing that is
happening is

now i have a secondary source of
information so in the dyad up until now
i my only source of information was the
physical environment if i was low
skilled like i think when

raphael was presenting like the orange
language the low skill low skill person
they had very little

uh they had a very low ability to
understand where they are in where they
are or are

in that environment and hence they were
not very successful in achieving their
goals

most of the time but the high skilled
the blue line which you know came down
really quickly uh

had a very accurate understanding of
where they are on the map so they could
quickly traverse to their goal

the thing that changed was by having
high social perceptiveness now i have
a secondary source of information about

the environment my primary skill is gone
like i cannot infer from you know the
chemicals but i can infer from my
partner's behavior so in that sense
there is additional information that is
helping the individual agent be a better
aif

right because now they have a source of
location of information that they can
trust and you know build accurately on
that said

that source of information is not
changing their own

weighting of their goals and that's why
there's no top down like there is no uh
as a collective we are not trying to
influence each other we are only taking
independent decisions

now assisted by a social environment
also and that that much is enough to see
collective uh outcomes emerge but if you
are looking for more complex scenarios
in which you are supposed to make a
decision of should we you know break
down across multiple tasks or what how
many of these people should do task a
and do task b or understand who is good
at what those kinds of things will
require more sophisticated machinery
so it is i would not say it is a
proper af

well that to that i would add my counter
find that uh if you define a proper af
model as just any markup blanket then
yes the dyads are
are um

markup blankets composed of the
where the internal states are the union

of the two agents and uh
yeah even the even the bigger collective
is just a very boring uh
agent
right the collective is an af agent it
has its uh its inputs and it has its
active state which is the null state
right but
you know proper there mean is it useful
to think of it of that of it in those
terms i guess it it depends right we
want to
we wanted to learn some salient facts
about the collective we didn't quite ask
that many questions about the diet as a
as its own unit but we could
so in the diet formation did you guys
play with like i know that you did the
strong and the weak and you had like the
strong one had strong environmental
preferences and you had some parameters
that you played with like and and the
weak agent didn't have any like was
really stupid about the environment but
like could lock on to the partner and
infer based on the partner so did you
ever play with parameters like
what if you make one agent um
you know weak in both and one agent
strong in both or or what what happens
like did you mess around with those at
all is it like
the scale of it like as you were tuning
to get like what was just right for you
uh
yes i think we're not can go into a lot
of detail on that uh we did learn one
thing very interesting which is uh you

could very easily replicate the
phenomenon of the blind
the blind leading the blind
where uh and we turned out it turned out
that if uh
if the strong agents was
also very socially perspective it was
very easy for that agent to lock into
the partners behavior and
try to learn trying to try to infer more
than was actually
uh actually legitimate from from what
the the partner was doing which was a
sports almost brownie in motion and that
actually you know
it can throw a poor agent into a
tailspin so
uh it is it is when you start looking at
these uh
at these
more like as i said less less rule
following behaviors
they are pretty sensitive to to
to kind of this balance of parameters
well that is exactly it i think
a fantastic question blue uh and rafael
has captured the essence of it is
there is a combination so like we had
the entire like i was doing robustness
things and like there's an entire space
of
what are the different combinations and
how do they pan out the most striking
example was
when the strong agent has a very strong
social perceptiveness also uh
they even if they've achieved their
target they start to move away from that

because they're you know trying to get something they're getting confused about their own state because they think that the other member is still trying to move towards their goal state so either there must be some other goal state or like this is me putting explanation on top of the behavior that we saw so like we did not see a stabilization coming up now what this tells us it tells us a very important thing that our model is not complete at all right it simply says that you need these two features but these two features are not enough to see a fairly sophisticated version of collective behavior in which there is uh error correction at that level and and from my perspective again and i know that uh raphael and i uh you know have been going at understanding active inference and how we define it and that's why i come out on the side of the collective is not completely an active insurance model yet because it does not have error correction uh and how do we get there like there is more who wants to be built into that uh how is it different from if you have both all the agents in your team are very good in their physical environment so so that's like having an all-star team we know sometimes all all-star teams fail can we you know bring up that kind of

that niche of behavior now all of these questions are unanswered and extremely exciting you know important areas to go into uh but all of that and that's what you know i come down and we started with norms and we thought that we should have this and we should have that but we boil down to okay what is the minimum thing that we need to say without this nothing works so the collective does not emerge even in minimal circumstances without these two features uh in play if i could give one comment on that and it's such an interesting tension that your two answers brought out where if the system is designed as an active inference agent it's a no-brainer yes it is and then this debate which we've seen play out you know increasingly across so many different areas okay it's a system that wasn't explicitly designed as an active inference agent whether it directly computationally emerges from a designed agent or whether it's a performer on a stage or some other social system of governance and then there's the realism question is it actually quote an active inference agent versus the instrumentalism question can we use this framework statistically to do inference and just like the distribution doesn't have to be perfectly normal in nature for the use of a t-test there's ways to

use active inference very usefully so
it's so cool to see like a sandbox for
exploring some of these fundamental
questions about collective behavior
which often times it feels like every
single species or every system they have
their own programming language or they
have their own norms or the edges are
transient so they go down
the rabbit hole with the math that way
instead of stable edges
so what's the framework that's going to
have the flexibility to actually cross
these different
bridges within the collective
intelligence or collective behavior area
so it's an awesome good question
and i think that it goes to the heart of
like local
i don't know if you guys haven't read um
the three body problem where they have
uh you know uh spoiler alerts there's uh
an alien alien species and you know
they're
uh they have this big computer that's
made up of soldiers with flags
uh you know
the soldiers with flags make up a
computer that's by definition it's a
computer because it was designed you
know they're they're following rules
that that caused it to to compute
uh certain functions right and you know
that's that's how we design computers
now
there may be
uh certain you know turing machines that
emerge that are that are uh

that we can even sometimes even prove
that they are uh universal sewing
machines
and that's pretty awesome right and you
know that's another way of saying that
they are computers and i think there's
the same uh same sort of uh question
going on here with uh with active
inference where
you know is it a markup blanket yeah you
can
and draw the the the lines around it
it's a mark of blanket is it's
um
to what extent it's
performing free energy minimization
depends on how you define the generative
density how you define the the
the the
internal state so you can calculate the
free energy functional and
and
and yada yada and the
proof is inputting right when you look
at it does it like does it actually look
like it's optimizing that that free
energy function sometimes the answer to
that is
really hard to get or only feasible to
get if you make some pretty uh
aggressive assumptions as you're saying
about uh mean field um
and so on so it's
it's at the heart there's uh there's an
epistemological presence
and almost an ontological one
i'm going to ask a question from the
chat this is from

dean

who wrote

is there a difference between a
self-actualization loop and what might
be described as a sense of
self-awareness

and what is the pursuit of shared goals
dependent on if common interpretation
and translation such as symbolic
manipulation is not generating a common
context thank you

that's an awesome question i don't
pretend to have any answers to it i'll
give you my uh

sort of opinion i'd also love to hear
from the others uh i think um my mental
model of this is hey uh these agents are
reasonable they only have
to the extent that you can say that they
have uh beliefs or opinions about
anything they have beliefs or opinions
about two things right their their uh
their own position and their
their partner's position

so

they

in particular they don't have and these
are literally just uh
uh numbers from 1 to 16 right so there's
two probability distributions
in particular they don't have anything
that we could
associate with
the self as a concept
right because they're only looking at
two positions right now i think there's
uh
definitely uh

it's it's not trivial to say well what
is uh does self-awareness mean a
symbolic sense of self or is it just the
the
epic phenomenon of having
beliefs about some aspect of itself
right
i'm not qualified to judge that but in
my opinion what the agent has
is
just information about a specific aspect
of itself
and
it feels like without the symbolic
without like having a sort of more
sophisticated causal model that has
opinions about this opinions about that
and opinions about how these other
things can act and one of those things
is the self which is representing is
like a pointer to pointer to self
i don't think that qualifies itself
awareness
awesome these are fun discussions blue
daniel you needed
oh it was on the stream but not agency
so blue oh
um i will uh
follow up to that question actually so
you have like this
self-actualizing loop with the agent and
then you have this like other act
actualizing loop right and i kind of
looked at it like your maybe your
self-actualizing loop is like your iq
and your other actualizing loop is like
your eq maybe like or or either way um
but but what i really thought about is

like you know we are by nature like such
incredibly selfish creatures and so like
it seems that you like maybe gave equal
weight in the model like can you maybe
dig into like

because in in reality we don't give
equal weight like we would give the self
loop way more priority than the other
loop
and was that a parameter that you played
with

yes like i have a lot more to say but
like rough i'll go first and then
that is yes

but if i go ahead please yeah no no i'm
just going to say yes i think the the
the
primary question that we were looking at
when we think about the these these
loops

if if you think about the
um

just the basic function of
of taking some input
from the

from the environment forming some uh
some

belief states that are just just really
like a collection of
of numbers and then uh
and then acting on it

uh that can represent pretty much
anything that's what we
were when we talk about a partner
actualization loop you can really just
think of it as two copies of the same
circuit one of them is
is

uh
wired
wired out into my own actions
and the other one is not
right so
the one that's wired out of my own
actions is the cell phone
self-actualization group just by virtue
of the fact that it's completing the the
it's the thing that's completing the the
markup blanket uh the the other one is
is wired out into nothing so it's just
generating predictions about my my
my partner's behaviors
but it's it has the same formal
structure internally to the
internally to the agent uh
now i think the the part where it gets
interesting is that uh consistency
requirement where we uh
uh we're talking about
like we have uh we have beliefs about
the about the
[Music]
the other and as as the
predictions are that we're generating
about their actions are related to those
beliefs but i think what we're asking is
really about uh what matters most to me
right is it about
uh am i paying more attention to my
actions and my goals or to to my
partner's actions and to
and to their goals right so i think what
the definition of matters
is really about what's influencing my
actions
that is primarily about my beliefs but

they are informed about my beliefs of
it's primarily about my beliefs of my
position but they're also informed by by
my beliefs about my partner's position
which is that process of triangulation
that we talked about and that's the
parameter the theory of my parameter
that we
were
we're squeezing out so
it is of course uh
you know it's very simplified but in
principle that's uh that's a range that
can go anywhere
i think we did play around even with uh
with having some uh
some
anti-social behavior uh programmed in uh
and you know
limited amount of interpretability
yeah i think uh blue very insightful
question again and uh
like i think we were also thinking about
as the self as being
iq because it is individual skill and
the partner actualization group being
more social intelligence because it is
about understanding the other person's
interaction
we sidestep that problem of
how do you weight one over the other by
choosing uh
you know the space
and again like i think this maps on very
well with your previous question that oh
what happens if both of them like if uh
an agent is good at both right we see
blind leading the blind which means that

as soon as you see that diversity in agents that can be really good at both or really bad at both you'll need some form of weighting mechanism that to what extent do i value information that i have

received from my primary skill versus i value information that i've received from my social environment and by social environment now as you can imagine if you have more than one partner you'll get you will also get conflict information because the partners might be moving differently because they are differently able right so at that time we cannot sidestep the question of how do i weight the social information coming in from different partners so there has to be a trust or a credibility index that i have to maintain internally so this is very close to uh concepts like a transactive memory system in management literature right where we are trying to set up that oh i know that uh raphael is an expert at skill a

and that's a credible source so even if i find a new team member who is equally skilled at uh you know uh the same skill okay i would more like i'm more likely to go to rafael for help when i need it because he has credibility he has some history so there is an index that i maintain internally a meta memory of mapping of who has what skills and how credible they are so like you cannot sidestep that question as we start building

and working towards bigger collectives
right so you are spot on with that and
that will start to happen even at the
simplistic level of this discrepancy can
be created between my primary uh
information source and my social
information source
we did not
like as like we
played around with oh let's give it
antisocial like they always think that
their physical thing is better
this led us to having conversation like
do they have meta memory which is like
do they have self awareness that they
are low skilled in the physical
environment as compared to their partner
right so this all started building more
complex things like in my other stream
of research i built those kinds of
agent-based models which are social
cognitive architectures that you give
them a memory they give them a meta
memory you give them meta attention you
give them a reasoning a way to map who
is working before how much do i trust
them so like and that's where these
ideas were flowing in from that all that
complexity we cannot sidestep that
complexity if we want to build
you know that sophisticated behavior
that was not the goal of the paper and
that's why we didn't like we just
removed all that like we just
sidestepped that complexity by saying
let's look at what is the minimum gains
that can be had by doing x or y
but the next step really if you are

trying to build a more sophisticated
behavior outcomes right not just
minimizing system level free energy
would be to poke in that direction
so brilliant question like i'm like this
is actively of how we've been thinking
and you know making the
model more complex and then simplifying
this is the exact tension and
interesting insights that you would get
there so very happy to uh you know if
you have more ideas on oh this would be
something that would be nice to pursue
i'm all yours

you know one one thought that that
brought to mind was when there's just a
differential equation model like some
kind of smooth line that's going to be
generated the model can't really fail
like you could fail to read in your data
set but it's not like the line's ever
going to break so
then people are wondering about how well
does it fit and they kind of focus on
the performance and then with
agent-based modeling it's always
understood that there's actually only a
few islands of parameters and of course
the choice of which parameters to
include
from which effective collective behavior
is going to arise like it's not just
enough to be an ant forager not every
foraging strategy works and the one that
works in the desert is not gonna work in
the jungle so it's actually just these
islands of stability and then you bring
up all these questions like well what's

the resilience of it to being
perturbed those are the the kinds of
questions that we care about with the
real world systems and it's almost like
the affordance from a scientific and a
modeling communication perspective of
having
a modeling approach like linear
regression that pretty much accepts any
kind of data set without really
questioning it it leads to different
questions that are being asked and it
leads to different uses of the model so
it's just so cool that those kinds of
questions are arising and i'm sure they
come up in interesting ways in your own
research so
we'll keep on yeah
yeah this is super exciting because uh
that's the direction that uh
sorry did i speak over you raphael
no no so this is
clearly as it has been communicated this
is super exciting to me because i think
uh a big question that we've not asked
yet is what is the goal of this kind of
a modelling like there's one way to look
at active inference can do x y z like
from my perspective the goal is okay how
can we have these social cognitive
models and active inference models go
hand in hand so that we understand
something about real teams and real
teams have you know as you know you
really elegantly put they have
non-linear behaviors they have islands
of places where they'll behave in a
structured manner and suddenly it breaks

because one parameter moved out of that island

to be able to get there we need some minimal machinery so i think that is the space where i live in when we uh you know

with anita and with chris riedel uh and i think i've not worked directly with david angle but like anita has pre-work with like that is the direction that we are pushing in in that space is like oh let's find this island of area where if

meta tension and meta meta memory is you know

reasonably developed or there's diversity within the team how does that where are the stable coordination zones there right right so that is one way but in aif at this point we did not even have the minimal machinery that i think we could understand and interpret so we started like what's the first building block that we need to build and the confession is i don't think i completely understand aif rules i think they're fluid like how do you define an aif like raphael was bringing up if it is only a mark of market then yes it is if it is this then it is that so this could be my shortcoming because i'm not red enough or it could be the field is actually really updating itself so so this is that's why this is exciting like let's build that but for me my eye is always on the like how do we pull it out into is this applicable like we don't want only equations that can

never be

wrong

in a matter of time you don't get

non-linear dynamics out of it directly

sorry yeah awesome thank you

yeah and then

100

agree and i think this uh

perhaps you you eluded that and i want

to pull it

uh pull it up it's

it's really about the

the scientific endeavor as a collective

active inference

uh behavior right if you think about how

uh

how scientific

modeling in the in the

kind of linear

paradigm

works it's about you know establishing

uh universal truths that are not

contingent

uh you know that

are you know contingent only on uh on a

very

uh

very specific very explicit set of of

assumptions that are knowable

and

within that you know you pull it out

uh you can explain it to an alien or you

can

pull it out and

write it into stone tablets and the next

civilization can read it and and so on

right and that's that's not what we're

doing here right i think we're feeling

our way as a as a collective towards a
more useful more um
more
interesting and applicable way to to
understand the real systems that are
around us and then to
maybe to learn something to help them
function better so that's that's
probably working at it
um i'll ask a quick question from the
chat because i think it applies and then
blue
so dean wrote where would the authors
see citizen learning organizational
structures
blind leading the blind or basic
research at scale
no it's not a buzzfeed article from 2027
it might be maybe you're seeing a
glimpse of the future
i don't know enough to answer brandon
you guys want to take a step
but now if you're muted yeah then
sorry yeah i was just responding to
dean's comment it's a such an insightful
comment there's like one hopes that this
basically scale it is it it is like it
brings up the
uh thought experiment that will
democracy thrive if
uh you don't have independent thinkers
right like so if it is blind reading the
blind that nobody is actually thinking
everybody is just based on secondary
information then it would look one way
and it would be the same
at the aif level it's the same behavior
or it is independently thinking and

basic research at scale and you get completely different outcomes in both cases so like

i i don't think i have good answers there but i think

very very legitimate question uh something that we can figure out i think that's that's model that's modelling and it relates to the discussion earlier about convergent collective behaviors versus divergent so whether it's like honeybees deciding on where to relocate or some other type of consensus protocol you want convergence but then you talked about an example where there was preference to be on separate niches and it's also important to have divergence like in discourse and in other areas and creativity so how to design that space is so critical

i i think i think uh

your spotlight but there is value in both of those see convergence is important to get the outcome like the bees have to decide on where to go finally but divergence is important to assess the accuracy of the information so to the extent

uh or rather to explore the space right like

accuracy and exploration of the space itself so to the extent it is blind leading the line i would say that is biased towards convergence which means

the system is opening itself up to not settling on sub sub optimal outcomes uh because everybody agrees very quickly on

what the solution should be uh lack of creativity that kind of thing if you are too divergent and you're not able to converge then

the system is opening itself to not getting the rewards because nobody's agreeing on anything so you need both and that's what i think the independent research and the blind linear brand are fantastic things i don't think can resolve the issue unless there is uh

one agent that is always understood to be they have the most accurate information always and the entire system understands that right like it could be like a small family universe's or whatever mom says is correct so all the children's are lined up because they are not trying to question they agree and trust that that is true and in that case convergence is great anyway

um and actually to connect that to some of the analytical components of active inference in the free energy minimization there's one term that's often framed as pragmatic value and then the other one is the exploratory the epistemic component you know and accuracy minus complexity these different framings that we've seen and so it's so cool that you described the need to have a balance of the converging and the diverging collective algorithms not worrying about whether it is a formal active inference at the higher scale but just some of the same

principles which is that you have to have some target seeking or signal detecting

aspect and then some aspect that prevents at the very best premature convergence

if not actually keeping options open to remain resilient so those are some of the the principles of collective systems that kind of go beyond even active inference alone and just it doesn't matter at that higher level exactly what it is and so really interesting stuff

yeah so if you look in the literature uh in the management so you could look at mega and even corrective and uh complex adapter systems from holly arrow like they have a book on groups as complex adapter systems and like one of the key things that you know in in a separate line of research i build on that is there are two functions that any group has to do one is efficiency the other is maintenance the idea of efficiency is that the resource that you have you have to deploy very well and get the maximum out of it the idea of maintenance is that oh you need to choose the right goals so that the system can survive right because if you're very efficient but you're going after the low rewarding things then the system will not thrive on the other hand if you're choosing the right goals but your wasting away the resources and not applying them efficiently the system

will not survive so those two things
together are functions of any group or
any collective so i think
again like for me it's like they're
looking at the same phenomena from two
different sides one is coming from
what's the mechanics of it how do we
achieve that so you look at all the
social social cognitive architecture
collective memory collective attention
collective reasoning the other is
like does this look like an aif and can
we model it mathematically without
worrying about what exactly is the
mechanism right so it can prove as a
ground truth to
this team has way more to achieve than
it is currently achieving because an
ideal aif model can be better performing
right and you're missing something so
there is some losses happening in the
processes in the groups right it could
be we're not understanding we're not
choosing the right strategy or we are
biased towards convergence whatever that
is right like that's something to be
diagnosed about the team and that
therein lies a practical application
like you have to figure out from here to
here what could be you know going wrong
so yeah all right
the thing that i want to say
i wanted to add that is
if you look at gaia theory uh and if you
shake it hard enough and it looks like a
dynamical system and you're
throwing the the uh ergodicity
assumption

yeah i know maybe from the
from the whole
whole system throughout it's like
billions of billions of years uh time
scale perspective and you can back out
and say yeah it's a single it's a single
active inference agent and therefore
there is one you can back back it into
like one big system goal for all time
that is only ever visible in retrospect
from uh from the perspective of like
some of the the time keepers in the
multiverse or whatever right uh we hear
simple agents uh try to do our part
don't have access to that and we have to
do with more proximate goals right these
are often going to be at odds these are
often going to turn out to be wrong and
there's
there's
always this tension between how you
construct uh
the goals are you constructing them for
for exploitation or for
for exploration at what at what level
how can we can in turn internalize
uh
internalize the work of other
uh systems around you to to get towards
more of a
uh more bang for the buck for everybody
and so on uh but it's all it's it's all
really imperfect and i think that's what
the the question trying to finally tie
it back to that question on
on citizen learning it's you know
from uh from a long run perspective
probably it's

to some definition it works out okay but
what okay means and uh you know in the
long term we're all dead
it's uh
the devil is definitely in the details
on that one
nice that reminds me a lot of bucky
fuller and humans role as sensemaker
locally and making it work for everybody
so it's really cool to see that kind of
come back to active inference through
design science and other areas so blue
so i have like a small technical
question and i'm just gonna load it up
followed by my big question so pranav i
am curious about your like social
science mapping and meta mapping and
memory and meta memory just like how
much compute power that takes like is it
is it monstrous but then i'll i'll add
another a couple couple more things so i
have done a little bit of a
agent-based modeling if you like need
somebody to interrogate you over
your model parameters i'm super good at
that so
i will harass you any time um and then
uh the real question that i have is
maybe because of the nature of
agent-based modeling maybe it's
you know you always have to define the
parameters of the system
and so we're looking to create a model
where essentially the rules are broken
right like when you make a collective
the collective has broken the rules it's
done what the individual agents it's not
collective behavior it's something new

right and so we don't know what that newness is going to be and so maybe agent-based modeling is is maybe the wrong tool like how do you allow in an agent-based model where you define all the parameters how do you allow for this like newness to emerge

again blue fantastic question i'm so happy to keep listening to your questions they're just very insightful uh

i i'm not sure if this is if i've understood the spirit of your question but i think based on what i understand i would say the use of agent based modeling is predicated on the goal that you have so then in a social science or sort of like social theory perspective uh we are building a transactive systems model of collective intelligence that is we are saying that collective intelligence emerges out of uh the formation and co-regulation of processes that involve collective memory collective attention and corrective reasoning so intelligence is an outcome of these three functions but these are not independent as in every person has that that is true but there is coordinated so collective memory which is called transactive memory in the literature emerges out of the skill based interactions that coordination that people do same thing with attention same thing so if we're building that

kind of a model

the role of an agent-based approach to

this kind of theorizing

is essentially

to say that okay i have this fantastic

narrative theory that people understand

are aware of each other's skills meta

memory right and they use that

information

to make

allocation choices that is if i have a

certain task and it requires help with x

skill who do i go to or who do i assign

this to in the team so like if that is

the theory

how do we know that this is enough this

is parsimony so this is the idea of

generative sufficiency right like are

these minimal processes enough to

explain the emergence of your collective

memory right like will a flock emerge if

each bird is each void is given these

three rules right so

and that kind of

research question or you know building

of that model is important to basically

give some credence to these are the

minimum social cognitive processes that

are important to give you this outcome

so we are not trying to code flocking we

are only giving individuals certain

capacities and the flock is emerging so

that this is minimally enough now is

that flock adaptive to

uh let's say a predator approaching them

maybe not because we've not told them

how to react to a predator right so in

that sense we now know there is a

boundary condition that with only x rules it can you know handle high levels of workload but it cannot handle high levels of uncertainty like those kinds of things so computationally how difficult or how complex the thing depends on how many features you are trying to build into it the model that i've you know developed for my dissertation and i'm you know presenting uh as i'm going on the market uh is fundamentally saying that at the simplistic level give each individual agent three capacities of having skills and meta memory understanding you know my team members their skills having attention or focus like ability to work on something and understand how other people are working on different things so like what priority that is how important it is for them which leads to the meta reasoning which is i have selfish motivations and i know that other people also have selfish motivations but is the task aligned with you know each other motivation so i should send this task to somebody for whom this is already aligned so they'll get to it faster and that will benefit the team it's like none of this is a centralized uh here's the leader who knows everything and is making the allocations right this is not top down command control this is very decentralized how will an intelligence at the collective

level emerge to the extent these members are able to figure these things out so that that's that's the building of the model that's you know headed towards that direction so i think what it becomes useful is under what circumstances the useful question to us is under what circumstances is meta reasoning or you know collective memory important so like this is the question that is very different from saying that or do we need this parameter or not and how computationally intensive it is the the answer lies in okay what's the environment what's the so like as you know daniel was pointing out there are islands of operation does this island in a phase space land in a place where these parameters are important or not so those are the kinds of questions you can answer with that and i think that would drive the choice of parameters i would not i would not try to say that or it should be a all encompassing model it should be the simplest model uh if anything i've tried to always say that we're trying to go to the simplest model that explains her part very clearly does that answer the question i i suspect it does not yeah so so um definitely like with flocking and like i've seen similar models of like fish schooling and and stuff like that um so we have that as a model

of maybe collective behavior versus
collective

intelligence or like has something
actually has something new
emerged in the flock or in the school of
fish

and so my my point is is that when you
just give these agents their individual
parameters how do you know when there's
another level like we're looking for
another level that does active inference
so so not like a yes to collective
intelligence you can even collective
intelligence and collect behavior can be
modeled with a group versus the
individual agents but when do you have
another level like in the hierarchy when
do you when do your cells become a
tissue when does your group of people
become you know a squad or a team or
something like this

so that that's what i don't know if
agent-based modeling is suited to to
determine or not

well that's a good question i think
it depends on

so it's partly a definitional issue
what do you mean by another level right
so intelligence the way it is defined at
least in that literature collective
intelligence is an ability to adapt to
changing environments so as the task
changes in its complexity that is
uncertainty is changing or the level of
knowledge interdependence or the level
of workload is changing over time the
team is able to seamlessly adapt its
internal coordination processes to

maintain performance so it is not just because the task has changed in its complexity so the environment is uh moving around just because the environment moved it should not be so specialized that it starts to fade it is able to adapt to it so if intelligence is defined that way to what extent can the you know the team regulate internally to maintain performance at the highest level possible then i would say yes it is emerging because the the diagram that you show right like essentially the plot that you show is a team that does not have these coordination mechanisms is failing like is dropping in performance anytime there is a perturbation while a team that has these mechanisms is like seeing a momentary dip but is able to maintain high levels of performance so it is it is a relative thing now if you change the composition the internal coordinations would be different because now it's a different set of people trying to coordinate but it would not invalidate the idea that a different set of people still have collective intelligence or not because it's given by given our internal resources how are we adapting to the changes like if you're able to track their so in that sense i think agent-based model does a fantastic job of showing you that we are able to see this emergent behavior of maintaining high level

performance uh or not right and that's
where you you go into the mechanisms
that this mechanism is essential for
this outcome versus not
does that help

it does it does but it's it's the
question of at what scale is it an
active inference agent right so like is
the team the active inference agent or
is it the each individuals is it like
their collective active inference versus
the team collective i can see daniel
like oh wait i was just gonna say that
yeah

to speak to the question of novelty
there's the question of generating
novelty and then responding to novelty
and in the case of generating novelty
especially for humans we haven't really
talked about stigma g niche modification
cumulative culture so

there's

many avenues from collective behavior to
more niche modifying algorithms that
are

maybe explaining some of the phenomena
that we are interested in here but then
this question about responding to
novelty and like how do you deal with a
threat that maybe hasn't been seen
before or a failure mode that's never
been observed or rare

it kind of just reminded me of
homeostasis which is generalizing across
specific situations to help you keep
target variables within a preferred
range
and then

instead of just reacting to changes
it makes one move towards anticipating
which is the whole area of variously
like cybernetics and signal processing
and so those features of responding and
anticipating and planning
agent-based modeling is a great place to
computationally and physically test that
and
clearly active inference at the very
least can be
sketched out over a lot of these systems
and then it will be a debate always like
about the details and the mechanisms
like you brought up but it's just to
have a framework that we can integrate
scales and just nest them cleanly
is at the very least a path towards
answering some of these questions if not
philosophically at least empirically
raphael

yeah

thank you yeah and i think about the
philosophical aspect also uh
as

backing down from
unless you're looking at the gaia theory
thing and you don't believe in such a
thing as uh you know

um

the

uh

selection multiverse

uh thing uh

so if you back down to like subsystems
of the big thing whatever the big thing
is

then you can back down into functions

right and you can use that as your
benchmark for for for what performance
or success or intelligence uh looks like
and deconstruct
uh down
and
match that up with you get from from the
from the
bottoms up from the bottom of you and
yeah i guess the the
uh the interesting question is
when and where does it actually
do the the bottom up and the top down
approach
line up
right
and
you know there's i think in general
there's no
there's no
answer there from uh from
from an ecological perspective i think
there's some some pretty clear
patterns of functional specialization
and then and uh
some of them are hierarchical some of
them are you know tend to be like almost
hierarchical or hierarchical with a
little little bit of splint sprinklings
of of
uh horizontal
conditions and maybe that's a that's a
useful you know philosophical approach
to to to look at things that from from
the instantiation at least from the
instantiation of oh the kinds of systems
that were
interesting interested in it's when does

a group of cells become
become a tissue but it's also
uh
what does it what are the conditions
that
that a thing needs to meet
uh to be
to be meeting the function of a tissue
and to be uh
worthy of being called the tissue in
that in that functional context right
and then you can talk about well what
what if that tissue is not made of of
cells but of like
nano nanobots or whatever
and so on
yeah i do yeah i don't think it is a it
is an either or type of question i think
it is more of
uh the two approaches of like coming at
it from a mechanistic perspective versus
coming at it from an active inference
perspective are complementary they're
trying to
look at the same phenomena but estimate
them differently
so the idea of uh i think then started
mentioning that
morphogenesis and morphostasis i think
that that's like when novelty happens
how do you respond to that novelty is
part of you know one term in the aif
equation and it is also a way of looking
at saying that oh will the system adapt
to it or not right so there's a
mechanism of how it adapts and then
there is the equation version of it if
it adapts well it would be reducing its

uh free energy right because that is what we are able to estimate i think both of them are very complimentary approaches what so one way i think i look at it is how do you know that the adaptation from a mechanics mechanistic shampoo that is happening is correct or wrong right is it maladaptive is it going to lead for a better outcome or is it going to lead to a worse outcome one way of looking at it is is that reducing free energy or is that increasing free energy of the system right now that is where you can marry the two things and forecast compliments that or there are certain adaptations because as an agent is sitting inside the system you know not knowing the bigger picture there is no way you know uh whether these you know uh variations that you're trying are good or bad like you need a system level out so i think that is how i think they're complimentary again i don't have good answers yet because that's the area of research that you're trying to figure out but i the fact that that is where it could go we could combine the two approaches to get more insight on the both the mechanism and the overall outcome this has been great any last thoughts or any time you're always welcome to join again and it's just super interesting to hear about we hope that people watching it

live and replay could think about some
of these questions but anyone's welcome
to give a last thought otherwise this
was just a great first appearance with
you both and we hope to see you again
yeah just thanks this was super fun to
read and to discuss and i think um
i think that the the
theory underlying the model is
hot and super on point and building it
out and flushing it out into a more
thoroughly um you know developed model
will be awesome
thank you yeah now i really appreciate
it uh thanks to to both of you guys for
for inviting us uh thanks of course to
everybody who's uh watching this
or reading transcripts or you know
connected to them to the
mind meld and you know getting an
instant download of all humanity's
knowledge 200 years from now
uh and yeah i mean
as i said i think the ultimate purpose
of all of this is to be useful and it's
an evolving conversation
so we're really curious to to
hear thoughts questions uh other ideas
that might pop up and
yeah i mean i don't know if your emails
are
are available but
if not you can
make them available you can reach out to
us on twitter or down to to the active
inference labs organizers and they can
connect us
yeah

yeah same here thank you both of you and
thank you to serval and dean for the
questions

uh we are i think both of us are
available on twitter or otherwise to
have questions would love to follow up
on certain things you know and see if we
can make interesting projects out of it
thank you folks this has been fun thank
you bye cheers bye
good times